

# YICHEN (ANNA) TU

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## EDUCATION

**Duke University, Pratt School of Engineering**, Durham, USA

**Expected May 2027**

Master of Engineering, Financial Technology, **GPA:4.0**

- **Relevant Coursework:** Programming (C++, Python), Software Engineering (C++, Python), Asset Pricing, Statistics, Quantitative Risk Management

**University of St Andrews**, St Andrews, Scotland

**September 2021 – May 2025**

Bachelor of Science (Honours) in Economics and Mathematics, **GPA:3.7**

- **Relevant Coursework:** Causal Inference, Time Series Analysis, Asset Pricing, Mathematical Modeling, Advanced Linear Algebra

**University of Virginia** (Non-degree, Exchange Student), Charlottesville, USA

**August 2023 – May 2024**

- **Relevant Coursework:** Real and Complex Analysis, Econometrics, Stochastic Processes
- **Technical:** C++, Python (pandas, numpy, scikit-learn), R Studio, SQL, MATLAB, Latex

## PROFESSIONAL EXPERIENCE

**Fortune SG Fund Management**

**Shanghai, China**

*Global Markets Intern (Investment Department)*

June 2025 – August 2025

- Built Python models to analyze ETF volatility and diversification risk, visualized factor contributions for risk reporting.
- Synthesized analysis for fund presentations on HuaAn Oil & Gas LOF (162411.SZ) by tracking NAV/price performance and linking movements to OPEC+ output and market news.
- Extracted key risk drivers from 10+ sell-side roadshows on HK/US equities and macro, informing portfolio assessments.

**Hwabao Securities**

**Shanghai, China**

*Mutual Fund Research Intern (Research and Innovation Department)*

June 2024 – August 2024

- Analyzed ~50,000 raw data points using multivariate regression and identified operational risk factors that significantly influenced process time, including the impact of returned data and differences across various stages of approval.
- Utilized Python to process index stock weights and factor exposure datasets for 9 indices, calculated factor exposures for ~1,000 data points per stock, and leveraged Excel to visualize key insights for leadership decision-making.
- Built Python data pipeline to prepare multi-quarter financial data for quantitative risk modeling and strategy evaluation.

## LEADERSHIP EXPERIENCE

**STEP Diving into Data 2024**, University of St Andrews

May 2024 – July 2024

- Conducted EDA and pre-model prep on ADHD datasets using R Studio (3,000+ rows; 7 vars incl. ADHD OR, metal exposure, year): source merges, missing-value handling, numeric parsing from strings, and categorical-to-numeric encoding for risk modeling.
- Employed multifactor regression analysis to examine environmental risk factors (e.g., metal exposure, soil concentration rates) influencing ADHD incidence.
- Led a team of 7, assigning tasks based on strengths, tracking team outputs, and coordinating progress meetings with mentors.

## PROJECTS AND RESEARCH EXPERIENCE

**Honours Dissertation – Standardization and IPW for Causal Inference**, University of St Andrews

September 2024 – May 2025

- Estimated public health risk of smoking on overweight incidence using BRFSS 2023 data (~400,000 obs), applying standardization and inverse probability weighting (IPW) in R Studio.
- Built and compared parametric and non-parametric methods to assess sensitivity of risk estimates under varying covariate sets and modeling assumptions.
- Found ATE estimates ranging from -5.5% (IPW) to -9.3% (standardization); highlighted model-driven uncertainty and implications for robustness in risk quantification.

**Research Assistant - Business School**, University of St Andrews

September 2024 – December 2024

- Merged two international datasets, Clarivate (1 TB, containing assignees, inventors, patents) and Crunchbase (smaller, with worker names and employment histories), using Stata's reclink2 command to assign match probability scores from 0 to 1; achieved 5-17% match accuracy with a minimum threshold of 0.6.
- Optimized large-scale Python fuzzy-matching pipeline (1 TB Clarivate + Crunchbase) to enhance entity-resolution accuracy.

**Mathematical Modeling and Simulation - Business School**, University of St Andrews

January 2023 - May 2023

- Applied Newton-Raphson and Runge-Kutta methods to solve equations and simulate population models; used Python to model single- and two-species populations, generating phase diagrams and steady-state analyses.
- Visualized model stability and dynamics using Matplotlib through function and phase diagram plots.
- Assessed model outcomes to verify the accuracy and convergence of numerical methods, gaining insight into population growth, predator-prey interactions, and system sustainability.